Explainability Fidelity and Deployment Cost in a Trustworthy Cervical Cytology Screening Pipeline

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Abstract. Vision-transformer explainability methods for medical image classification are typically evaluated against one baseline at a time, on whichever backbone a given paper happens to train, without asking whether the choice of explainability method matters more or less than the choice of uncertainty-quantification method for the same underlying trustworthiness goal. We evaluate three explainability methods (relevancy propagation, attention rollout, and Grad-CAM adapted for vision transformers) on a cervical cytology classifier, using both behavioral faithfulness (deletion/insertion) and a real, expert-annotated localization ground truth (nucleus boundaries from the SIPaKMeD dataset) rather than a derived proxy. Relevancy propagation wins cleanly on all three metrics. We then ask a system-level question our own prior uncertainty-quantification comparison could not answer alone: does composing two independently-useful uncertainty signals – temperature-scaled confidence and conformal prediction-set size – into one abstention rule help, hurt, or do nothing? The answer is conditional: composition helps exactly where the better individual signal was already struggling (under severe distribution shift) and hurts where it was already excellent (in-distribution and under mild shift). Finally, we measure what trustworthiness costs in practice: resource-efficiency claims for lightweight vision transformers, evaluated by parameter count and FLOPs alone, do not reliably predict real CPU inference latency, and returning an explanation alongside a prediction costs roughly 3.3× a bare forward pass on CPU. Taken together, these results argue that explainability method choice is more reliably consequential than uncertainty-signal composition strategy, and that both claims require direct measurement rather than architectural or metric-count inference.

Keywords: Explainable AI · Vision transformers · Selective prediction · Conformal prediction · Cervical cytology · Resource efficiency

1 Introduction

Cervical cancer screening in specialist-scarce settings is a plausible target for lightweight, on-device AI assistance, but a screening tool that fails silently – misclassifying confidently under conditions its training data didn't cover – is

more dangerous than one that fails and says so. Prior work established that a lightweight hybrid vision transformer (EfficientViT-M0) matches or exceeds much larger CNN and full-size ViT baselines on in-distribution cervical cytology accuracy, and that this same backbone can collapse to near-chance accuracy under cross-dataset distribution shift with no static architectural property predicting which models fail safely. A companion study asked whether any of four uncertainty-quantification methods' native confidence signals correctly flag that collapse, and found a genuinely mixed answer: MC Dropout's entropy signal is the only one that substantially improves shift-failure detection on the severe shift condition, while a deep ensemble – the best method in-distribution and under mild shift – does not transfer that advantage to the severe case.

This paper asks the analogous question for explainability – does the choice of explainability method matter as much, and in the same condition-dependent way? – and then asks a genuinely different question neither prior study could answer: given that no single component (backbone, uncertainty method) is reliably best across all conditions, does *composing* two different native uncertainty signals into one system help, or does composition just average away whichever signal was better? Finally, since a lightweight-architecture claim is only as good as its measured deployment cost, we report the actual CPU latency and memory cost of the full pipeline, not an estimate from parameter count.

Our contributions are threefold. First, a three-way explainability comparison (relevancy propagation vs. attention rollout vs. Grad-CAM) on deletion/insertion faithfulness *and* real expert-annotated nucleus-localization ground truth, where relevancy propagation wins unambiguously on all three metrics – a cleaner result than either companion study's uncertainty-method comparisons produced. Second, the first system-level test of whether combining two uncertainty signals for selective prediction helps: a conditional answer (helps under severe shift, hurts in-distribution and under mild shift) that extends the "no single best method" finding from individual methods to method combinations. Third, a measured, not estimated, resource-efficiency and deployment-cost analysis: parameter count and FLOPs do not reliably predict CPU latency across a 12-model registry, and an explanation costs $\sim$3.3$\times$ a bare forward pass.

2 Related Work

Relevancy propagation for transformers [1] extends attention rollout [2] by weighting each transformer block's attention matrix with its gradient with respect to the target class before the same rollout aggregation, making the resulting relevance map class-discriminative rather than a fixed structural property of attention. Grad-CAM [3] was originally designed for CNN feature maps and has been adapted to transformers by targeting a block's token-wise activations instead of a convolutional feature map; we identify and document a specific failure mode of this adaptation for CLS-token-pooled ViTs. MC Dropout [4], deep ensembles [5], and evidential deep learning [6] are the three uncertainty-quantification methods our companion study compares against temperature-scaled softmax; conformal

prediction's finite-sample coverage guarantee is the calibration method this paper's selective-prediction composition experiment builds on. Deletion/insertion faithfulness [7] is the behavioral-faithfulness metric used throughout. Directly adjacent to this paper's own scope: the author's own prior work combining a stacking ensemble with explainable AI for cervical cancer diagnosis [10], which this paper differs from by evaluating explainability methods against a real expert-annotated localization ground truth and behavioral faithfulness metrics rather than an ensemble-accuracy framing; and the author's "MaizeFormerX" lightweight-ViT-plus-explainability design for plant disease diagnosis [11], motivating this paper's own choice to hold efficiency and explainability as joint, not competing, design goals. Two further papers from the same author sit in adjacent clinical-AI territory: a semi-supervised retinal disease recognition system [12] and an explainable multimodal AI framework with reinforcement learning for post-surgical clinical decision support [13], both establishing a continuing research program in trustworthy, explainable clinical AI of which this paper's cervical cytology system is one instance.

3 Methods

3.1 Backbone

The winning in-distribution backbone from our prior efficiency benchmark, EfficientViT-M0, uses a hybrid convolutional + cascaded-group-attention design. Relevancy propagation and attention rollout are both defined for standard multi-head self-attention transformer blocks; EfficientViT-M0 has no established propagation rule for either method. Rather than force an ad hoc extension onto an architecture these methods were not designed for, this paper uses **DeiT-Tiny** [9] – a standard `timm VisionTransformer` achieving 0.9606 balanced accuracy on the same SIPaKMeD in-distribution task in our prior benchmark (vs. EfficientViT-M0's 0.9654) – retrained fresh for this paper.

3.2 Explainability methods

All three methods produce a relevance map over the model's 14×14 patch grid (DeiT-Tiny, 224px input, 16px patches), upsampled to image resolution. **Attention rollout**: raw softmax attention, averaged over heads, chained across all 12 blocks via the standard rollout recurrence. Class-agnostic. **Relevancy propagation**: identical rollout recurrence, but each block's contribution is $\mathrm{ReLU}(\nabla_{\mathrm{attn}} \odot \mathrm{attn})$, averaged over heads, where the gradient is with respect to the model's own predicted class's logit. Class-discriminative. **Grad-CAM (ViT-adapted)**: gradient-weighted channel importance applied to a transformer block's token activations rather than a CNN's convolutional feature map. *Implementation note*: the target layer must be the *second-to-last* transformer block, not the last. DeiT-Tiny pools only the CLS token for classification, so no layer downstream of the final block mixes information back into patch-token positions – verified directly

by inspecting gradients: every patch token's gradient at the last block's output was exactly zero, while the second-to-last block's patch-token gradients vary normally.

3.3 Faithfulness and localization metrics

Deletion/insertion AUC [7]: patches are ranked by a method's own relevance score and progressively replaced with (deletion) or revealed from (insertion) a heavily Gaussian-blurred version of the image, tracking the model's predicted-class probability at each step; the area under the resulting probability-vs-fraction curve is reported (lower is better for deletion, higher is better for insertion). **Nucleus concentration ratio**: SIPaKMeD ships real, expert-marked nucleus boundary polygons per cell; we persist these in crop-local pixel coordinates and compute the fraction of a heatmap's total relevance mass falling inside the nucleus region, divided by the nucleus's share of image area. A ratio of 1.0 is exactly chance level; higher values mean a heatmap concentrates relevance in the nucleus more than chance would predict – clinically the correct place to concentrate.

3.4 Calibration, conformal prediction, and the combined selective-prediction signal

Temperature scaling [8] is fit on a held-out calibration split disjoint from both training and test data. Conformal prediction sets are built via a resampling-averaged split-conformal threshold. The combined selective-prediction signal rank-normalizes temperature-scaled max-softmax confidence and (inverted) conformal prediction-set size independently, then averages their ranks.

3.5 Efficiency and latency measurement

Parameters and FLOPs (via `torch.utils.flop_counter.FlopCounterMode`) are architecture properties independent of trained weight values, so measured without any training. CPU latency and memory (isolated per model in its own subprocess) are measured with untrained weights for the same reason. The "cost of an explanation" is measured as a full forward+backward+rollout pass for relevancy propagation's own predicted-class explanation, against a bare forward pass, both on CPU.

4 Experiments

4.1 Explainability results

Relevancy propagation wins every comparison, and by a wide margin on nucleus concentration specifically (2.71× chance vs. attention rollout's 1.18× – barely above chance). We attribute this to a mechanistic difference rather than measurement noise: attention rollout is class-agnostic, an inherent disadvantage on an inherently class-discriminative faithfulness test, while relevancy propagation is both class-discriminative *and* integrates the gradient-weighted signal across all twelve transformer blocks.

Table 1. Explainability method comparison, 50 SIPaKMeD test images, all matched to a real nucleus polygon.

Method	Deletion AUC↓	Insertion AUC↑	Nucleus conc.↑ (1.0 = chance)
Relevancy propagation	**0.734**	**0.970**	**2.714**
Grad-CAM (ViT-adapted)	0.818	0.877	1.472
Attention rollout	0.825	0.936	1.177

4.2 System-integration results

Two proposed sub-claims about safe composition hold by construction given this project's post-hoc-attachment architecture, and were verified rather than assumed: temperature scaling changed **zero** argmax predictions on the calibration split, and explainability computation is a read-only analysis that cannot touch model weights.

Table 2. Selective-prediction risk-coverage AUC (lower is better): confidence alone vs. conformal set size alone vs. combined.

Condition	Confidence only	Set size only	Combined
In-distribution	**0.0050**	0.0834	0.0173
Herlev (moderate shift)	**0.1190**	0.3281	0.1793
Mendeley LBC (severe shift)	0.3244	0.4395	**0.2855**

Composition is not a uniform win. In distribution and under moderate shift, confidence alone is already the best individual signal, and combining it with the noisier discrete conformal-set-size signal dilutes it. Under severe shift, however, the combined signal is the best of the three, beating confidence alone. This mirrors the shape of our companion uncertainty-method finding (no single choice wins everywhere), now extended to combinations of signals.

4.3 Resource efficiency and deployment robustness

Across a 12-model registry, EfficientNet-Lite0 (3.38M params, 0.77G FLOPs) has more than 4× ResNet-50's CPU latency (137.7ms vs. 32.0ms) despite ResNet-50 having roughly 10× the FLOPs. Even EfficientViT-M0, the overall winner on params+FLOPs+accuracy, is not the fastest model in wall-clock CPU terms: DeiT-Tiny (15.4ms) and LeViT-128 (17.5ms) are both faster despite 10–14× more FLOPs. EfficientViT-M0, evaluated under synthetic JPEG compression, Gaussian blur, and brightness/contrast shift, degrades gracefully under all three, but Gaussian blur is by far the most damaging – costing 32.8 points of balanced accuracy at its most severe tested level, roughly double JPEG's worst case (15.5 points) or brightness/contrast's (18.1 points). A synthetic OOD sanity check

shows mean confidence dropping from 0.985 (real in-distribution images) to 0.743 (OOD) – real and meaningful, but far above the 0.20 chance level for this 5-class task. On CPU, a bare DeiT-Tiny forward pass has a median latency of 49.6ms; a prediction that also returns a relevancy-propagation explanation costs 164.1ms – about 3.3× the bare-prediction cost.

5 Discussion

Two threads run through this study. First, explainability method choice produced the cleanest, most unambiguous result so far – relevancy propagation wins every metric, with no condition-dependent flip – while uncertainty method choice and uncertainty-signal composition both produced conditional, no-single-winner results. This asymmetry suggests that for this task and backbone family, picking the right explainability method is a more reliably beneficial engineering decision than picking (or combining) the right uncertainty-quantification method, where the honest answer is "it depends on the operating condition." Second, every efficiency and robustness claim required direct measurement to be trusted: FLOP count did not predict CPU latency; and OOD confidence dropped only partially, not to the clean near-chance level a reader might assume from an "OOD detection" framing.

6 Limitations

Backbone mismatch: the explainability and system-integration results use DeiT-Tiny, not the efficiency benchmark's overall winner, EfficientViT-M0, since relevancy propagation is not defined for the latter; whether these results transfer to a hybrid conv+attention backbone is untested. Single-institution, small-scale data (SIPaKMeD, ∼4,000 images). CPU latency measured on one shared consumer laptop, not dedicated hardware; a ∼30% run-to-run swing was observed for the same model, though relative ordering across models was stable. OOD sanity check uses a small, synthetic sample (25 images) rather than a natural out-of-domain dataset. No clinical or expert-review validation of explanation localization fidelity.

7 Conclusion

Relevancy propagation is a clearly better explainability method than attention rollout or Grad-CAM for this cervical cytology classifier, on both behavioral faithfulness and real expert-annotated localization ground truth – the most unambiguous positive result in this line of work. Composing two uncertainty signals for selective prediction, by contrast, helps only under the specific condition where the better individual signal was already failing. Resource-efficiency and robustness claims for lightweight vision transformers require direct measurement in both directions: FLOP-efficient does not mean latency-efficient, and "confidence drops on OOD input" does not mean "confidence reaches a safe abstention level."

References

1. Chefer, H., Gur, S., Wolf, L.: Transformer interpretability beyond attention visualization. In: CVPR (2021)
2. Abnar, S., Zuidema, W.: Quantifying attention flow in transformers. In: ACL (2020)
3. Selvaraju, R.R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., Batra, D.: Grad-CAM: Visual explanations from deep networks via gradient-based localization. In: ICCV (2017)
4. Gal, Y., Ghahramani, Z.: Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. In: ICML (2016)
5. Lakshminarayanan, B., Pritzel, A., Blundell, C.: Simple and scalable predictive uncertainty estimation using deep ensembles. In: NeurIPS (2017)
6. Sensoy, M., Kaplan, L., Kandemir, M.: Evidential deep learning to quantify classification uncertainty. In: NeurIPS (2018)
7. Petsiuk, V., Das, A., Saenko, K.: RISE: Randomized input sampling for explanation of black-box models. In: BMVC (2018)
8. Guo, C., Pleiss, G., Sun, Y., Weinberger, K.Q.: On calibration of modern neural networks. In: ICML (2017)
9. Touvron, H., Cord, M., Douze, M., Massa, F., Sablayrolles, A., Jégou, H.: Training data-efficient image transformers & distillation through attention. In: ICML (2021)
10. Mohiuddin, A.B., Rahman, M.M., et al.: Accelerated and accurate cervical cancer diagnosis using a novel stacking ensemble method with explainable AI. In: International Conference on Quantum Photonics, Artificial Intelligence (2025)
11. Rahman, M.M., et al.: MaizeFormerX: a lightweight vision transformer with cross-scale attention for explainable maize leaf disease diagnosis. Scientific Reports (2026)
12. Gony, M.N., Shuvra, S.M.K., Poudel, R., Dhakal, K., Rahman, M.M., et al.: Dual-view semi-supervised retinal disease recognition via chromatic structural decomposition and cross-view contrastive learning. In: International Conference on Power, Electronics, Communications (2026)
13. Ahmed, M., Ahmed, F., Mow, S.M., Taha, P.A., Barua, S., Rahman, M.M., et al.: An explainable multimodal AI framework with reinforcement learning for post-surgical clinical decision support. medRxiv (2026)